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Extension of the method of characteristics for the design of supersonic stators in two-phase turbines

Two-phase turbines present a pathway to higher efficiency in energy-conversion technologies, including low-temperature heat-to-power cycles, heat pump and refrigeration systems, and liquefaction processes. Achieving high aerodynamic performance requires careful consideration of two-phase flows within the turbine. Due to the low speed of sound in two-phase mixtures, convergent-divergent nozzles play a critical role in enabling supersonic expansion in the stator blades. While the method of characteristics is well established for single-phase problems, its extension to two-phase turbine blades remains unexplored. To address this gap, we propose a new method for designing stator blades for two-phase flow conditions, ensuring both high efficiency and uniform exit flow. The divergent section of the nozzle is designed by generalizing the method of characteristics to two-phase flows, assuming thermodynamic equilibrium between the phases. The resulting nozzle geometry is integrated into a blade parameterization framework, enabling the generation of complete supersonic stator blade profiles. The method of characteristics is verified against two-dimensional fluid dynamic simulations employing a barotropic model, and the performance of the stator blades is examined under design and off-design conditions. The results demonstrate the suitability of the approach for designing high-performance stator blades for two-phase turbines and establish it as a robust, computationally efficient alternative to more complex shape-optimization techniques. Overall, this work represents the first extension of the method of characteristics to two-phase flows, paving the way for improved turbine designs across energy conversion systems.

Keywords: organic Rankine cycle, computational fluid dynamics, speed of sound, homogeneous equilibrium model, barotropic model, blade parametrization.

1 Introduction

The increasing demand for high-efficiency energy conversion technologies has raised interest in two-phase turbines. Many low-grade heat sources consist of sensible heat, which is poorly matched by the isothermal evaporation process of conventional Organic Rankine Cycle (ORC) power systems [1]. This mismatch creates a pinch point at the start of the evaporator that prevents the full utilization of the heat source [2]. In contrast, partial-evaporation and trilateral cycles can enhance power production by expanding a liquid–gas two-phase mixture in the turbine, which enables a better match between the evaporator temperature profiles and ultimately increases the net power output by 20 % to 60 % [3]. Moreover, in heat pump, refrigeration, and liquefaction systems, substituting conventional Joule–Thomson throttling devices with two-phase turbines can enhance the thermodynamic efficiency by recovering the expansion work while increasing the liquid content at the exit of the expansion [4,5]. As a result, the coefficient of performance of heat pump and refrigeration systems can increase by 10 % to 50 % depending on the working fluid and operating conditions [6], and the liquid yield in liquefaction processes can rise by 2 % to 5 % [7].

Despite their potential advantages, the design and operation of two-phase turbo-expanders remains a challenge due to the complex flow physics and the multidisciplinary nature of the problem [8,9]. The simultaneous presence of liquid and vapor phases gives rise to complex phenomena, including interfacial mass, momentum, and heat transfer, as well as transitions through various flow

regimes, which strongly influence the aerodynamic behavior, mechanical integrity, and potential blade erosion [10]. Addressing these challenges requires the development of innovative turboexpander architectures, advanced numerical modeling, and a deeper understanding of the two-phase flow physics within turbomachinery to guide the design process.

Research on two-phase turbines began in the early 1970s, driven by the pursuit of higher energy conversion efficiencies and the exploration of alternative energy sources following the 1973 oil crisis [11,12]. Since then, a variety of expander technologies have been investigated, including axial and radial inflow turbines as well as screw expanders [6]. Among these configurations, impulse axial turbines have shown the greatest potential due to their high efficiency, compactness, and operational reliability, and have therefore become the main focus of technological development [6,12,13]. This turbine type consists of a stator followed by an axial rotor. The working fluid is expanded and accelerated through the stator, undergoing phase change (flashing) as the pressure decreases. The resulting high-velocity jets are then deflected by the rotor blades at almost constant pressure, exchanging angular momentum and producing shaft power. Due to the low speed of sound of two-phase mixtures [14], supersonic conditions can occur even at moderate pressure ratios and jet velocities [12], requiring converging-diverging stator blades resembling the profiles of de Laval nozzles. Designing high-efficiency stators is challenging due to choked flow at the throat, strong density and velocity gradients in the diverging section, and the presence of complex shock-wave interactions [15]. Additionally, as the stator blades accommodate almost all the pressure drop and phase change while accelerating the flow to supersonic speeds, it is arguably the most critical com-

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ponent of impulse two-phase turbines.

Several approaches have been proposed for designing the geometry of two-phase nozzles. Many rely on one-dimensional space-marching [16,17] or time-marching [18] schemes, which are effective tools for estimating the area ratio required to achieve a target exit Mach number. However, these methods inherently assume that the flow is uniform across each cross-section of the nozzle, a simplification that averages out the inherently two-dimensional nature of supersonic expansions. In reality, the flow downstream of the throat is shaped by Prandtl–Meyer expansion waves and their reflections, which generate cross-passage variations in pressure, Mach number, and flow angle. As a consequence, one-dimensional models cannot resolve this internal wave pattern and are therefore limited in their ability to design nozzle geometries that aim to achieve highly uniform exit conditions, which is essential to reduce unsteadiness and aerodynamic losses in the downstream rotor. More advanced techniques, such as surrogate-based shape optimization frameworks, which couples computational fluid dynamics (CFD) with numerical optimization, have been applied to both single-phase [19,20] and two-phase flows [21,22]. Although such CFD-based optimization approaches can account for both design and off-design performance to generate robust nozzle geometries, their high computational cost makes them less suited for the preliminary design and rapid assessment of supersonic stator configurations.

As an alternative, the Method of Characteristics (MoC) has been widely adopted for designing the supersonic section of convergent-divergent nozzles [23]. Its formulation enables the direct integration of the governing equations along characteristic lines, ensuring smooth, shock-free expansions [24]. Beyond nozzle design, the MoC has also been successfully applied to the shaping of supersonic turbine stator and rotor blades, including high-supersonic axial turbine stages [25] and transition ducts for supersonic inlet turbines in rotating-detonation engines [26]. The design procedure begins with the definition of the sonic line at the throat, where the flow reaches critical conditions. From this boundary, the nozzle contour is obtained by integrating the governing equations across the characteristic network through the solution of the compatibility and characteristic equations. This process involves solving a system of Ordinary Differential Equations (ODEs), complemented by an appropriate Equation of State (EoS) describing the thermodynamic behavior of the fluid.

Recent studies have extended the classical MoC to real-gas conditions, revealing significant deviations from ideal-gas behavior, particularly for high-molecular-weight fluids [27]. Accurately representing the thermophysical properties of the working fluid is essential, as they strongly influence the slope of the Prandtl–Meyer expansion waves and, consequently, the wall shape generated by the MoC. Deviations in the real-gas speed of sound, which may even lead to non-monotonic Mach number distributions along the nozzle [28], can substantially increase the required area ratio and normalized nozzle length to achieve a given exit Mach number [29]. If these effects are not captured, the MoC will produce sub-optimal nozzle shapes that do not achieve an adapted expansion. Specifically, an excessive area ratio leads to over-expansion and the formation of shock waves that generate entropy and reduce efficiency, while an insufficient area ratio produces additional expansion waves downstream of the exit that reduce the flow angle.

Although the MoC offers clear advantages for designing high-performance blade geometries at low computational cost, it has not yet been extended to two-phase flow conditions. One possible reason is the lack of a unique expression for the speed of sound under two-phase flow conditions, which is needed to determine the slope of the characteristic lines. As illustrated in Fig. 1, the speed of sound of two-phase mixtures depends on the degree of mechanical, thermal, and chemical-potential equilibrium between phases. As revealed by relaxation analyses of the two-phase flow equations [30], several limiting forms of the mixture speed of sound arise depending on the level of interfacial coupling between phases. In the homogeneous frozen model (HFM), where the gas and liquid phases evolve independently without volume, heat, or mass

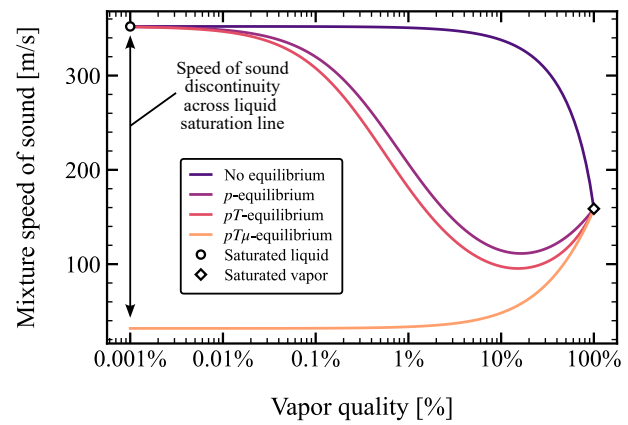


Fig. 1 Mixture speed of sound as a function of vapor quality under different equilibrium assumptions for cyclopentane at a saturation temperature of 200 °C, calculated using the expressions derived in [30]. Under full thermodynamic equilibrium, the speed of sound exhibits a discontinuous drop upon entering the two-phase region, which is especially pronounced at the liquid saturation line and much weaker near saturated vapor.

transfer, the speed of sound is the mass-weighted average of the individual phase speeds. By contrast, when the phases are allowed to exchange volume to reach pressure equilibrium, or volume and heat to reach thermo-mechanical equilibrium, the speed of sound has a continuous, non-monotonic variation between the liquid and vapor values. Finally, under full thermodynamic equilibrium, also known as the homogeneous equilibrium model (HEM), the mixture sound speed exhibits a distinct behavior: it becomes substantially lower than the value of either phase and presents a discontinuous drop upon entering the two-phase region, which is especially pronounced at the liquid saturation line and much weaker for the vapor line (see Fig. 1). To the best of the authors' knowledge, no extension of the MoC that considers the behavior of the speed of sound in the two-phase region has been proposed to date, and as a result, its influence on the geometry of nozzles and stator blades remains unknown.

To address this gap, we introduce the first MoC formulation for two-phase flow conditions and apply it to the design of supersonic stator blades. To this end, we generalize an existing real-gas implementation of the MoC by incorporating the homogeneous equilibrium speed of sound model, enabling the integration of characteristic lines for gas-liquid mixtures evolving under full thermodynamic equilibrium. The MoC framework is used to generate the divergent section of nozzles that produce uniform exit flow conditions, which are then embedded into a new parametric stator geometry model to generate full blade shapes based on engineering design parameters such as throat opening, blade metal angles, and passage width distribution. The capabilities of the proposed approach are demonstrated by designing the stator blades of a two-phase turbine for an exemplary cyclopentane partial-evaporation ORC system selected from the literature [31]. The flow fields predicted by the MoC are compared with CFD results obtained using a barotropic two-phase model. Additionally, an analysis of the flow field is performed at design and off-design conditions to identify the primary flow structures and loss mechanisms, providing insights into the aerodynamics of two-phase supersonic stators. To the best of our knowledge, this work presents the first MoC-based parametrization tool and CFD analysis of stator blades, providing both a practical design tool and new insights into the aerodynamics of two-phase turbines.

The rest of the paper is structured as follows. Sections 2.1–2.3 introduce the implemented MoC, the parametrization strategy for

the stator design, and the barotropic flow model, together with the numerical setup adopted for the CFD simulations. Sections 3.2 and 3.3 present the validation and verification of the MoC and the barotropic flow model. Section 4.1 examines the influence of the total inlet conditions on two-phase nozzle design. Section 4.2 discusses the design and off-design performance of a representative supersonic stator. Finally, Section 5 summarizes the main findings and outlines the key conclusions of this study.

2 Methods

2.1 Method of Characteristics. The governing equations for a two-dimensional, irrotational flow for a compressible fluid are [24,32]:

$$\left(u^2 - a^2\right) u_x + \left(u^2 - a^2\right) v_y + 2uvu_y - \delta \frac{a^2 v}{y} = 0 \quad (1)$$

$$u_y - v_x = 0 \quad (2)$$

where (u, v) are the cartesian components of the velocity, a is the speed of sound, (x, y) are the spatial coordinates, the subscripts indicate partial derivatives, and δ is a constant parameter set to unity or zero for planar and axisymmetric problems, respectively. For a supersonic flow condition, the gas dynamics equation, Eq. (1), becomes hyperbolic, and the MoC represents a common approach to translate the above Partial Differential Equation (PDE) system into an Ordinary Differential Equation (ODE) system, with a so-called compatibility equation:

$$\left(u^2 - a^2\right) du_{\pm} + \left[2uv - \left(u^2 - a^2\right) \lambda_{\pm}\right] dv_{\pm} - \delta \frac{a^2 v}{y} dx_{\pm} = 0 \quad (3)$$

By simplifying it and considering a planar nozzle, it becomes:

$$\left(u^2 - a^2\right) du_{\pm} + \left(v^2 - a^2\right) \lambda_{\pm}^{-1} dv_{\pm} = 0 \quad (4)$$

The characteristic lines themselves are described by

$$\frac{dy}{dx} = \tan(\theta \pm \alpha) = \lambda_{\pm}, \quad (5)$$

along which the compatibility equation is solved. The subscript $\pm$ denotes the right- and left-running characteristic lines, θ is the local flow angle and α is the Mach angle defined as $\sin^{-1}(1/\text{Ma})$. The MoC completes when the target Mach number is reached at the outlet of the nozzle. Conversely, if only the outlet pressure is known, the corresponding outlet Mach number can be obtained from the following set of equations:

$$\begin{aligned} a_{\text{out}} &= a_{\text{out}}(p_{\text{out}}, s_0), & h_{\text{out}} &= h_{\text{out}}(p_{\text{out}}, s_0), \\ v_{\text{out}} &= [2(h_0 - h_{\text{out}})]^{1/2}, & \text{Ma}_{\text{out}} &= \frac{v_{\text{out}}}{a_{\text{out}}}. \end{aligned} \quad (6)$$

Eq. (3) is closed by applying an Equation of State to compute the thermodynamic state of the flow. By assuming an adiabatic and isentropic expansion, the speed of sound can be related to the inlet stagnation conditions as:

$$\begin{aligned} h &= h_0 - (u^2 + v^2)/2 \\ a &= a(h, s_0) \\ s_0 &= s_0(P_0, T_0) \end{aligned} \quad (7)$$

where h_0 and h are the total and static enthalpies, and s_0 is the stagnation entropy. The mixture properties required for the computation of the MoC and for determining the thermodynamic state

at each intersection of the characteristic lines are evaluated under the HEM assumption, following the formulation by Flåtten and Lund [30]. In this approach, both phases are assumed to be in thermal and mechanical equilibrium, sharing a common pressure, temperature, and velocity. The mass-specific thermodynamic properties—such as internal energy, enthalpy, and entropy—are computed as mass-weighted averages of the corresponding saturated liquid and vapor quantities. For a generic mass-specific property ψ , the mass-weighted average is given by:

$$\psi = y_{\ell} \psi_{\ell} + y_v \psi_v \quad (8)$$

where y_{ℓ} and y_v are the mass fractions of liquid and vapor, respectively. By contrast, density and viscosity are computed as volume-weighted averages of the liquid and vapor values:

$$\phi = \alpha_{\ell} \phi_{\ell} + \alpha_v \phi_v \quad (9)$$

where ϕ is a generic volume-weighted property and α_{ℓ} and α_v are the volumetric fractions of liquid and vapor. The mass and volume fractions of liquid and vapor are related to the density of the mixture according to:

$$\alpha_{\ell} = \left(\frac{\rho}{\rho_{\ell}}\right) y_{\ell}, \quad \alpha_v = \left(\frac{\rho}{\rho_v}\right) y_v \quad (10)$$

Finally, the speed of sound is determined as:

$$\frac{1}{\rho a^2} = \frac{\alpha_{\ell}}{\rho_{\ell} a_{\ell}^2} + \frac{\alpha_v}{\rho_v a_v^2} + T \left[\frac{\alpha_{\ell} \rho_{\ell}}{c_{p,\ell}} \left(\frac{ds_{\ell}}{dp}\right)_{\text{sat}}^2 + \frac{\alpha_v \rho_v}{c_{p,v}} \left(\frac{ds_v}{dp}\right)_{\text{sat}}^2 \right] \quad (11)$$

where a_{ℓ} and a_v are the speeds of sound of the saturated liquid and vapor, $c_{p,\ell}$ and $c_{p,v}$ are the heat capacities of the saturated liquid and vapor, and $\left(\frac{ds_{\ell}}{dp}\right)_{\text{sat}}$ and $\left(\frac{ds_v}{dp}\right)_{\text{sat}}$ are the slopes of the liquid and vapor saturation lines in the entropy-pressure plane. The thermodynamic state is evaluated through the Helmholtz Equation of State (HEOS) implemented in CoolProp [33], which provides higher accuracy than cubic equations of state for both thermodynamic properties and their derivatives. While various formulations exist for calculating the speed of sound of a two-phase mixture (see Fig. 1), the HEM is expected to provide an accurate representation of the flow physics for the operating conditions considered here. This is because the presence of both phases at the inlet, combined with the high Mach and Reynolds numbers, is expected to promote intense mixing and interfacial exchange, resulting in small non-equilibrium effects along the nozzle. However, experimental evidence confirming this conjecture is still needed.

The numerical implementation of the MoC begins at the sonic line, whose shape is determined using the procedure of Sauer [34]. This method, based on a small-perturbation technique, is directly applicable to a two-phase mixture in thermodynamic equilibrium by properly evaluating the thermophysical properties under the HEM assumption. The nozzle geometry is then constructed in two stages. First, the kernel region is generated, where the main flow expansion occurs. It is followed by the turning section, where the streamlines are smoothly aligned to achieve a uniform exit flow. The kernel region is defined using a circular arc and terminates when the flow reaches the design Mach number. The turning section is obtained by enforcing mass conservation within the cells of the characteristic-line grid until the velocity component orthogonal to the axis of the exit flow is zero.

2.2 Stator parametrization. The diverging nozzle generated via the MoC is incorporated into a blade parameterization framework for supersonic stator design, as shown in Fig. 2. To reduce the number of points from the MoC, the nozzle contour is approximated using a cubic B-Spline, enforcing a maximum deviation

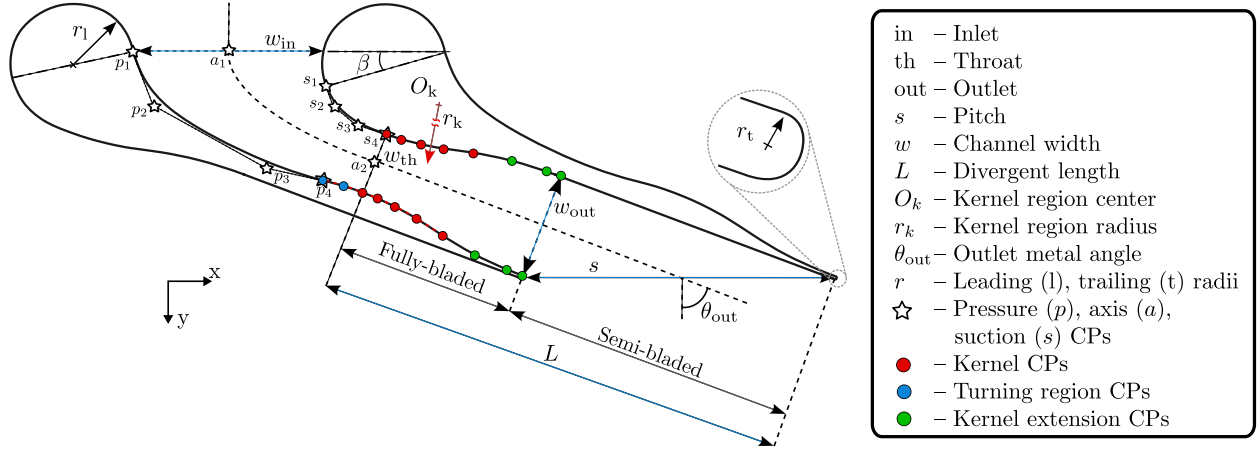


Fig. 2 Parametrization of the stator blades.

of 0.05 mm from the original geometry. The control points are mirrored with respect to the nozzle axis to obtain a symmetric divergent section. The resulting control points are shown with red and green markers, representing the kernel and turning regions of the MoC design, respectively.

In the MoC design, the divergent axis starts at point a_2 , which also represents the origin of the Cartesian reference frame for the MoC implementation. Point a_1 is positioned at coordinates $[-L/3, L/8]$, with L denoting the length of the divergent section, to ensure a gradual flow turning. A counterclockwise rotation of the metal angle at the outlet, θ_{out} , transforms the MoC coordinates into the turbine reference frame. The pitch can be calculated from the outlet height obtained from the MoC and θ_{out} as

$$s = \frac{w_{out} + 2r_t}{\cos \theta_{out}} \quad (12)$$

In the stator frame of reference, the convergent suction and pressure sides are modeled using two separate Bézier curves of degree 3, defined by the control points s_1-s_4 and p_1-p_4 , respectively.

On the pressure side, the first control point p_1 is defined based on a_1 , which lies on the symmetry axis of the passage, and the inlet width w_{in} , an input parameter. The circular arc representing the kernel region in the MoC is slightly extended into the convergent portion, ensuring a well-defined throat and continuous curvature, which mitigates flow separation in high-acceleration regions. The resulting control points, shown in blue in Fig. 2, include p_4 , which coincides with the end of the pressure side. The intermediate points p_2 and p_3 are positioned to preserve local tangency, at distances $L_p/4$ from p_1 and p_4 , respectively, where L_p is the distance between p_1 and p_4 .

The suction side is generated analogously, starting from s_1 , obtained by translating a_1 along the x -axis by half of h_{in} and rotating it by an angle β . The tangent at a_1 is further rotated by $\beta_{in} = 20^\circ$ relative to the turbine y -axis to promote smoother flow turning at the inlet. The last control point, s_4 , coincides with the starting point of the MoC at the throat. Intermediate points s_2 and s_3 are placed to preserve tangency, at distances $L_s/3$ from s_1 and s_4 , respectively, where L_s is the distance between s_1 and s_4 .

The leading and trailing edges are modeled as semicircular arcs: the leading edge connects s_1 and p_1 , while the trailing edge is defined with a 1 mm diameter. The profile is closed by a straight segment. Excluding the closing line, the complete geometry is represented by a single cubic B-Spline curve respecting a certain tolerance with respect to the original blade shape. The geometrical parameters of the obtained stator design for the reference case study are listed in Table 1.

2.3 Computational setup. The case study examined in this work includes both inviscid and viscous simulations performed in

Table 1 Stator geometrical parameters.

Variable	Value	Unit
Throat width, w_{th}	20.00	mm
Inlet channel width, w_{in}	70.00	mm
Outlet channel width, w_{out}	33.20	mm
Trailing edge radius, r_t	0.50	mm
Outlet metal angle, θ_{out}	70.00	°

Ansys Fluent [35] for two-dimensional geometries. Inviscid simulations solve the Euler equations, which are suitable for code verification but become insufficient when viscous effects play a significant role, such as in boundary-layer development, stator cascade efficiency assessment, or shock-boundary layer interactions. To capture these phenomena, the compressible Reynolds-Averaged Navier-Stokes (RANS) equations are solved using the two-equation SST $k-\omega$ turbulence model.

For both viscous and inviscid cases, spatial discretization is carried out using a second-order upwind scheme for the continuity, momentum, and turbulence transport equations. Convergence is reached when the root-mean-square (RMS) residuals of all transport equations fall below 10^{-6} and the mass imbalance between the inlet and outlet sections stabilizes below 10^{-6} .

Fluid properties in the CFD simulations are evaluated using two complementary approaches. For single-phase flows, Real Gas Property (RGP) tables are employed when CFD results serve as a verification for the barotropic model. These tables are generated using the same HEOS formulation applied in the MoC and barotropic model codes to ensure thermodynamic consistency and eliminate potential discrepancies from the default material database of Ansys Fluent. For two-phase flow simulations, a barotropic equilibrium model is employed, as described in the following section.

The inlet boundary conditions for single-phase cases are imposed in terms of total pressure and total temperature, (P_0, T_0) . For cases starting in the two-phase region, the thermodynamic state is defined from the pair (P_0, Q_0) , where Q_0 denotes the total inlet quality. In Fluent, only P_0 is imposed at the inlet for the two-phase cases, since the energy equation is not solved under the barotropic flow assumption. The inlet turbulence intensity and viscosity ratio were set to 5 % and 10, respectively, representative of low turbulence levels at the nozzle inlet. These parameters have no significant impact on the solution, as the inlet flow is characterized by a very low velocity [36]. At the outlet, the static pressure is imposed for all cases. For the nozzle simulations, only half of the domain was modeled by applying a symmetry boundary condition at the axis. For the stator simulations, the domain was limited to a

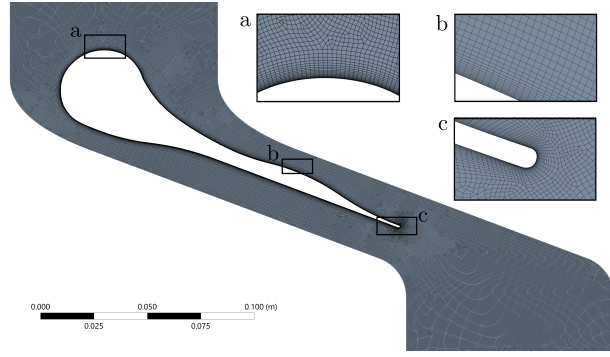


Fig. 3 Stator mesh with three zoom regions: (a) leading edge, (b) throat, and (c) trailing edge.

single flow passage, with periodic boundary conditions applied on the side boundaries.

For the viscous simulations, inflation layers are generated with a first-layer thickness chosen to ensure maximum values of y^+ below 1 in single-phase conditions and below 3 in two-phase conditions. Smooth walls are assumed throughout the study. A grid independence study was conducted using three meshes for each case: 20, 40, and 80 cells at the nozzle throat. The deviation between the 40-cell and 80-cell results was less than 0.15 % for both mass-averaged outlet velocity and mass flow rate. The 80-cell mesh was nonetheless chosen for the simulations due to the low computational time. The stator mesh used is shown in Fig. 3.

All simulations were performed on a personal computer equipped with a 13th Gen Intel Core i9-13950HX processor (24 cores). The barotropic CFD simulations of the stator blades were performed using 8 cores, requiring less than 200 seconds under design conditions and less than 300 seconds under off-design conditions. The MoC designs, for an outlet Mach number of approximately 1.5 as considered in this study, required approximately 150 seconds when executed on a single core. The current version of the MoC is implemented as single-threaded interpreted Python code, and significant speed-ups could be achieved through parallelization and just-in-time compilation.

During validation and verification of the results, the Normalized Root Mean Square Error (NRMSE) is used, defined as:

$$\text{NRMSE} = \frac{\|Y_{\text{sim}} - Y_{\text{ref}}\|_2}{\|Y_{\text{ref}}\|_2}, \quad (13)$$

where Y represents the quantity of interest.

2.4 Barotropic property modeling. The barotropic fluid model is adopted to evaluate thermophysical fluid properties in the simulations involving two-phase flows. Being part of the HEM family, the barotropic model ensures thermomechanical equilibrium between the phases, neglecting the possibility of modeling non-equilibrium heat and mass transfer [37]. The HEM formulation of the barotropic model is consistent with the one applied for the MoC implementation. From a computational and robustness perspective, the barotropic flow model offers a major advantage over other HEM formulations, since the energy equation does not need to be solved.

The barotropic model assumes that fluid properties depend solely on pressure, thereby simplifying the formulation compared to other models. In reality, fluid properties also depend on entropy, and therefore, an assumption on entropy variation is required across the expansion process. In this work, a barotropic process with a polytropic efficiency of one is adopted, implying that the entropy remains constant at the stagnation value, $s = s_0$. This approximation is expected to have a minor impact on the CFD results, as high irreversibilities do not occur in the simulations performed. However, this assumption does not imply that the flow is isentropic;

Table 2 Boundary conditions for the validation with experimental data from Spinelli et al.[39].

Case	P_0 (bar)	T_0 (K)	Z_0 (-)
1	9.02	542.15	0.65
2	4.58	520.15	0.83

rather, it suggests that viscous dissipation and thermal effects do not influence the precomputed thermophysical properties, which are evaluated along an isentropic line. Mechanical dissipation is included through the momentum equation, closed using an SST $k-\omega$ turbulence model, while entropy generation does not affect the evaluation of the barotropic thermodynamic properties, which are computed under the isentropic assumption. Although an iterative procedure could be implemented to include a finite polytropic efficiency, literature has shown that the resulting differences are negligible [38]. In this context, the isentropic efficiency of an expanding process can be defined as:

$$\eta_{\text{is}} = \frac{h_{0,\text{in}} - h_{\text{out}}}{h_{0,\text{in}} - h_{\text{out, is}}} = \frac{v_{\text{out}}^2}{v_{\text{out, is}}^2}, \quad (14)$$

where $v_{\text{out, is}}$ is the outlet velocity for an isentropic expansion, and v_{out} is the mass-averaged CFD value.

The mixture properties required to solve the flow equations, such as the density ρ_m and viscosity μ_m , and the quantities required for the post-processing, such as quality Q and speed of sound a , are computed with Eqs. 8-11. All properties are implemented in Ansys Fluent using polynomial functions, obtained by fitting the data computed from CoolProp [33] or REFPROP [40] using the barotropy package [41].

3 Verifications and validations

3.1 CFD model validation. The CFD model is validated against the experimental data of Spinelli et al. [39]. Their measurements concern the single-phase expansion of MDM in convergent-divergent nozzles and exhibit significant real-gas effects. The nozzle considered in this study is designed for an outlet Mach number of 2.0. The boundary conditions for both cases are reported in Table 2, indicating total compressibility factors Z_0 significantly deviating from the ideal gas value of unity. The nozzle includes a small geometric step at the throat to generate a weak shock wave, enhancing Schlieren visualization during the experimental campaign [39]. Two sets of two-dimensional viscous CFD simulations are performed: one using RGP tables to import the thermo-physical properties of MDM, and one employing the barotropic model described earlier.

Figure 4 compares the simulation results with the experimental measurements, showing the (a) static pressure and (b) compressibility factor distributions along the nozzle axis for the two cases considered. The barotropic model predictions match well those based on the RGP tables, with a NRMSE of 0.34 % and 0.01 % for the pressure and compressibility factor distributions for Case 1. Neglecting the energy equation in the barotropic model does not introduce a source of error in these cases, as viscous dissipation has a negligible effect on the fluid thermodynamic state and the expansion is nearly isentropic. The shock generated by the backward-facing step at the throat is clearly captured in both solutions, with a sharp disturbance visible at a normalized axial position of about 0.75. A weaker reflected wave, formed after the shock meets the nozzle wall and turns back toward the centerline, can also be seen around a normalized position of roughly 3. The comparison with experimental data demonstrates that the fluid properties are accurately captured even in the presence of significant real gas effects. The barotropic simulations have a NRMSE with respect to the experimental data for Case 1 of 2.74 % and 0.92 % for pressure and

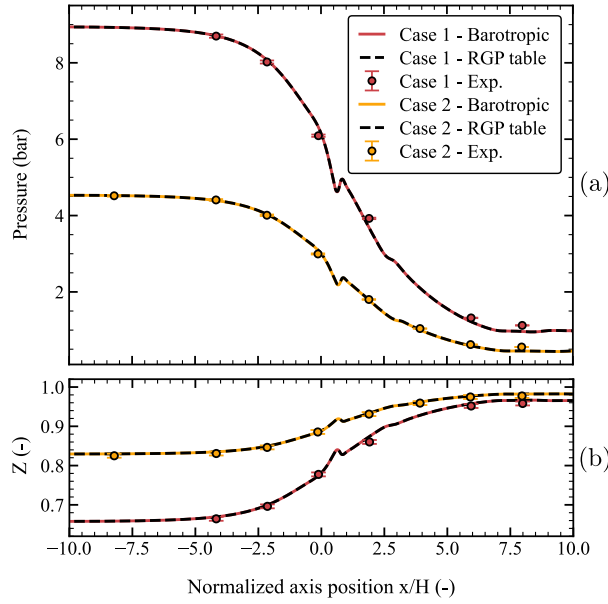


Fig. 4 Validation of the barotropic model for real gas with the experiments from Spinelli et al.[39]. (a) Pressure and (b) compressibility factor distributions along the nozzle axis.

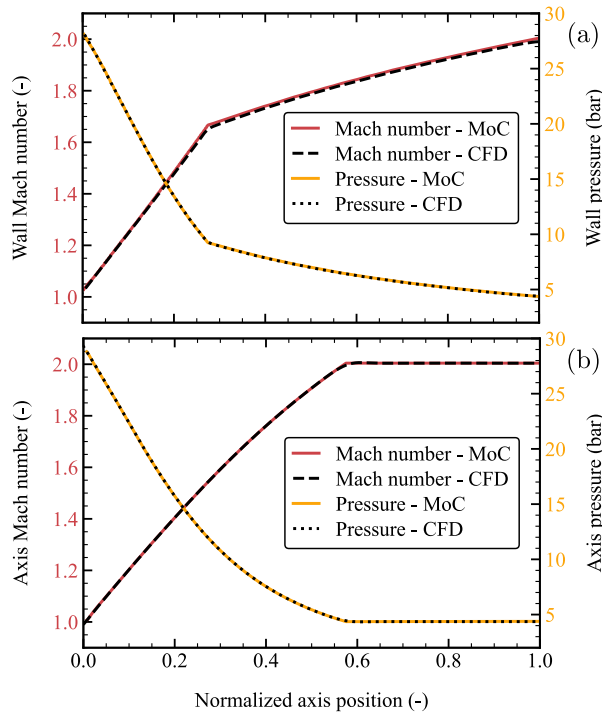


Fig. 5 Validation of the method of characteristics for single-phase flows. (a) Wall and (b) axis Mach and pressure distributions.

compressibility factor, respectively. For Case 2, the NRMSE are 1.88 % and 0.38 % for pressure and compressibility factor.

3.2 Method of Characteristics verification: single-phase.

The MoC code is initially verified using a single-phase expansion with toluene as the working fluid. The stagnation inlet conditions are chosen close to the critical point to enhance the level of non-ideality, specifically $P_0 = 1.01P_{cr}$ and $T_0 = 1.01T_{cr}$, where P_{cr} and T_{cr} denote the critical pressure and temperature, respectively. The

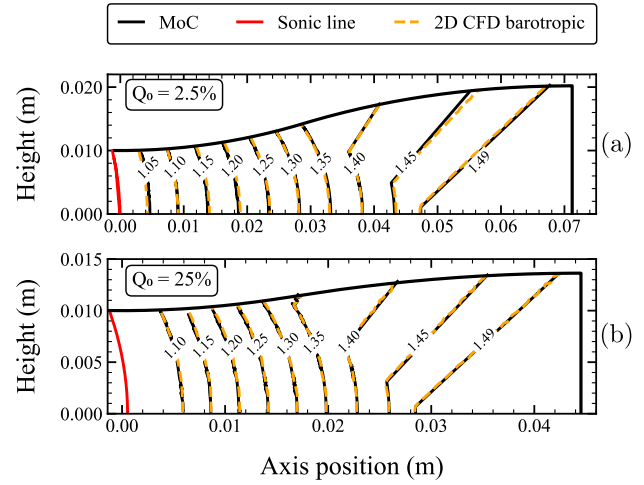


Fig. 6 Comparison between the Mach lines obtained from the MoC design and two-dimensional CFD simulations employing the barotropic model for total inlet qualities of (a) 2.5 % and (b) 25 %.

MoC is employed to generate the divergent portion of the nozzle, while a conventional convergent section is added to ensure a smooth convergence of the simulations. Two-dimensional inviscid CFD simulations are used for verification, consistent with the inviscid MoC assumption. RGP tables provide the thermodynamic properties, allowing both approaches to rely on the same equation of state.

The Mach number and pressure distributions along both the nozzle axis and wall are shown in Fig. 5. The plots begin at the throat, which is located at the origin of the x -axis. The NRMSE between MoC and CFD predictions is 0.70 % and 0.15 % for the wall Mach and pressure, while 0.10 % and 0.08 % for the axis Mach and pressure. The small errors demonstrate the accurate implementation of the MoC, achieving the target outlet Mach number of 2.0. Furthermore, a relative error of 0.024 % in the mass flow rate predicted by the MoC compared to the CFD results confirms the accuracy of the design tool.

3.3 Method of Characteristics verification: two-phase.

A representative case from the literature is selected to demonstrate the application of the MoC under relevant two-phase operating conditions. The selected case is the turbine of the partial-evaporation ORC system analyzed by Tammone et al. [31]. This system corresponds to a geothermal source with a mass flow rate of 100 kg/s and a temperature of 100 °C, and the cycle operating conditions were obtained by techno-economic optimization aimed at minimizing the levelized cost of electricity. Following their results, a total inlet pressure of 2.513 bar is imposed, and two inlet vapor qualities are considered: 2.5 % and 25 %. The working fluid is Cyclopentane, a widely used hydrocarbon in both refrigeration and ORC applications.

The nozzle generated using the MoC is extended with a convergent section and simulated with two-dimensional CFD analysis employing the barotropic model for property calculation. The inviscid simulations are run to be consistent with the MoC approach assumption during the verification process. This method enables a direct comparison between the CFD and MoC results, particularly along the nozzle wall.

The iso-Mach number contours reported in Fig. 6 show excellent agreement, confirming the accuracy of the MoC in predicting two-phase expansion behavior. The target Mach number of 1.50 set for the MoC design is perfectly achieved at the nozzle exit. Additionally, the critical mass flow rates for both cases have relative errors of less than 0.05 % compared to the CFD results, demon-

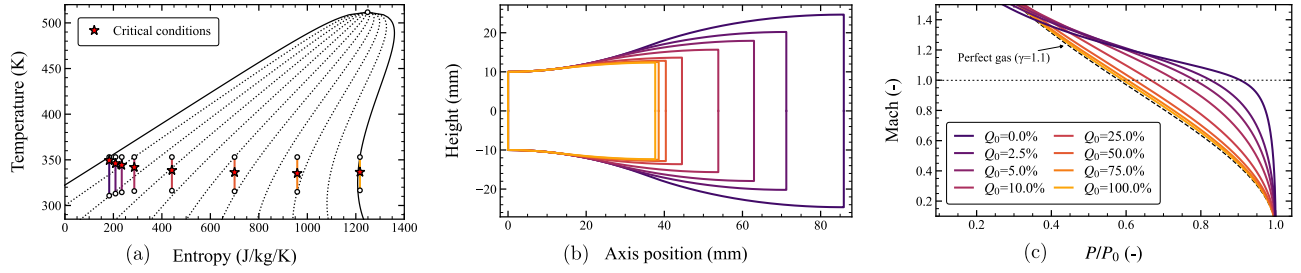


Fig. 7 Sensitivity analysis of total inlet quality variations for a fixed nozzle exit Mach number of 1.50. (a) Expansion processes on the Cyclopentane T - s diagram; (b) corresponding nozzle geometries obtained using the MoC; (c) isentropic Mach number evolution as a function of the normalized pressure P/P_0 .

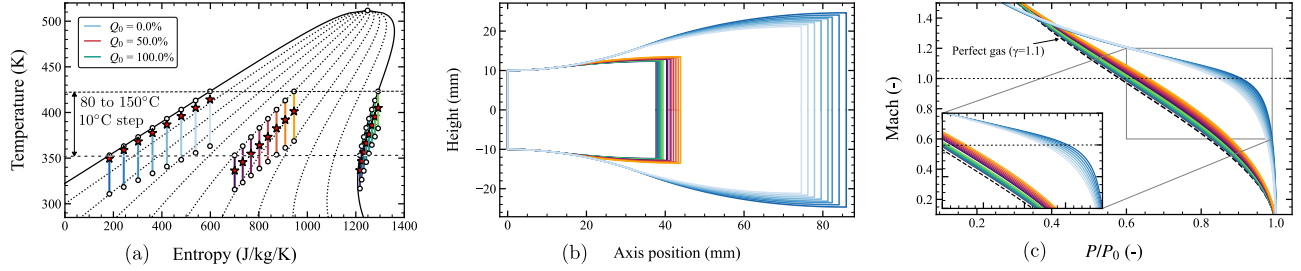


Fig. 8 Sensitivity analysis of total inlet quality and temperature variations for a fixed nozzle exit Mach number of 1.50: (a) Expansion paths on the Cyclopentane T - s diagram; (b) corresponding nozzle geometries designed using the MoC; (c) isentropic Mach number evolution as a function of the normalized pressure P/P_0 .

strating the robustness and predictive capability of the proposed formulation for two-phase nozzle flows.

4 Results and discussion

4.1 Two-phase supersonic nozzles design. This section examines how variations in the stagnation inlet conditions affect the nozzle geometries obtained with the MoC. The study by Tammone et al. [31] is taken again as a reference case for fluid and inlet conditions.

4.1.1 Sensitivity analysis on inlet quality. The first sensitivity analysis examines the effect of varying inlet total quality for the same total inlet pressure condition of 2.51 bar. The exit Mach number for the MoC design is set to 1.5 for all the cases. Fig. 7 (a) shows the set of operating conditions considered on the Cyclopentane T - s diagram, spanning inlet total qualities from 0 % to 100 %. The 100 % case corresponds to a single-phase expansion, due to the dry nature of the fluid. The red markers indicate the sonic states that are reached isentropically, which correspond to the initial point for the divergent section of the MoC-based nozzle design.

For low inlet qualities, the critical condition lies close to the inlet state due to the substantially reduced speed of sound in two-phase mixtures characterized by low vapor qualities (Fig. 1). The resulting nozzle geometries are reported in Fig. 7 (b). A pronounced sensitivity of the nozzle length and throat-to-exit area ratio to the inlet quality is observed. The nozzle for an inlet quality of 0 % is more than twice as long as the one obtained for a 100 % inlet quality, primarily due to the significant increase in the exit-to-throat area ratio.

The Mach number distribution as a function of normalized pressure is shown in Fig. 7 (c). For low inlet qualities, the curves display steep gradients near the sonic condition, reflecting the sharp reduction in speed of sound. As the inlet quality increases, these trends gradually flatten, shifting a larger fraction of the pressure drop to the subsonic portion of the nozzle. This trend reflects the

Table 3 Boundary conditions and key performance indicators for the two-phase supersonic stator simulations.

Quantity	Over-expanded	Design	Under-expanded	Unit
$P_{0,in}$	2.51	2.51	2.51	bar
$Q_{0,in}$	2.50	2.50	2.50	%
P_{out}	1.20	0.95	0.70	bar
v_{out}	71.21	90.71	111.19	m/s
$v_{out,is}$	74.37	91.96	113.61	m/s
α_{out}	72.34	69.98	63.64	°
Ma_{out}	1.26	1.38	1.49	—
η_{is}	91.68	97.30	95.78	%

behaviour also seen in the T - s diagram, where lower inlet qualities lead to larger pressure drops occurring in the divergent region. The single-phase expansion displays the closest behaviour to the perfect-gas trend described by

$$\frac{P}{P_0} = \left(1 + \frac{\gamma-1}{2} Ma^2\right)^{-\frac{\gamma}{\gamma-1}}, \quad (15)$$

and is also reported in Fig. 7 (c).

Although the nozzle contours in Fig. 7 (b) appear relatively uniform, the T - s diagrams show that the nozzle size increases more rapidly when the expansions occur near the saturated-liquid line, where the density ratio changes abruptly. In contrast, for higher inlet qualities, much larger changes in quality are needed to produce similar variations in nozzle geometry. This indicates that the key factor controlling the nozzle shape is the density ratio between the phases, which directly affects the void fraction of the mixture.

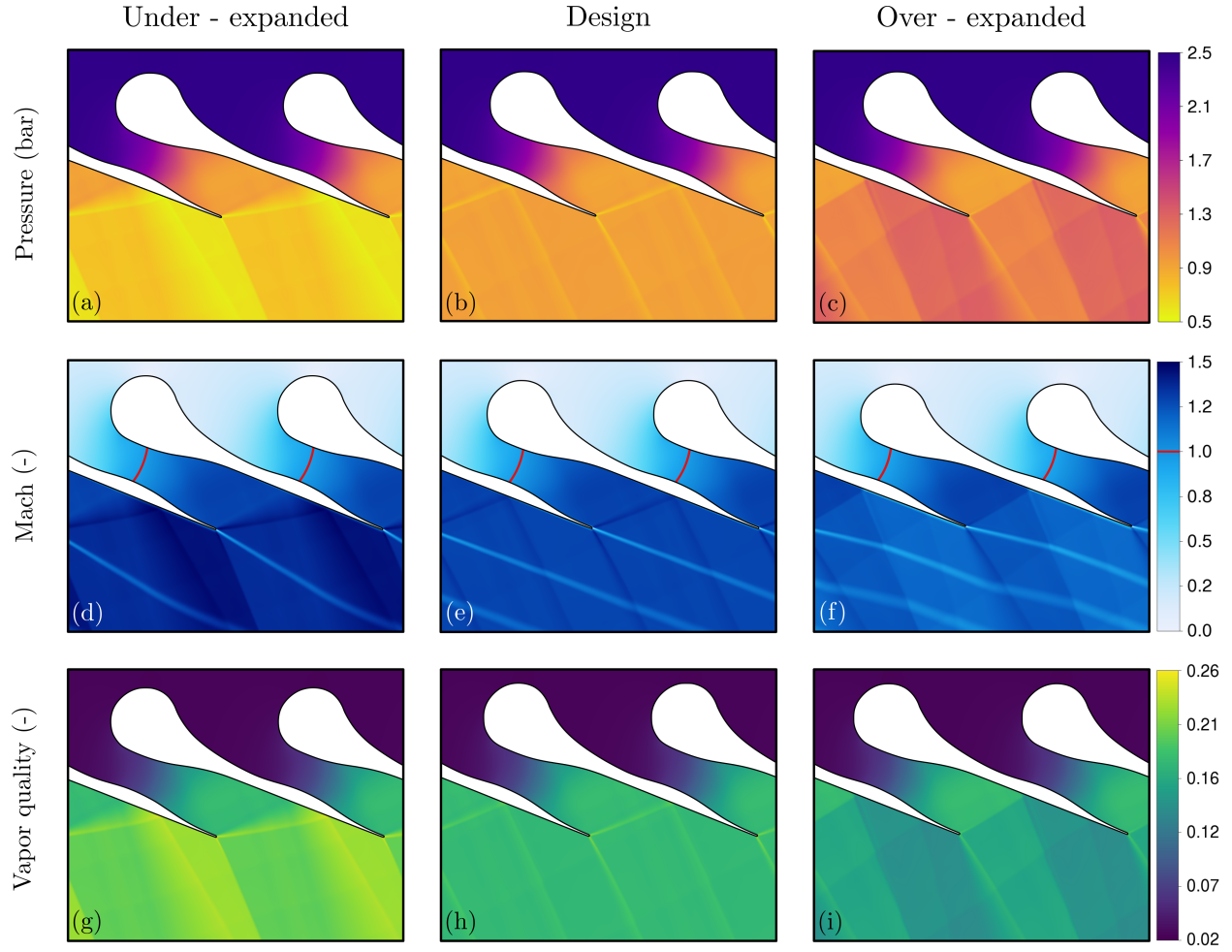


Fig. 9 Outlet pressure is varied to obtain three flow conditions: under-expanded ($P_{out} < P_{design}$, first column), ideally expanded ($P_{out} = P_{design}$, second column), and over-expanded ($P_{out} > P_{design}$, third column). The top row (Figs. 8a–8c) shows pressure contours, the central row (Figs. 8d–8f) shows Mach contours, and the bottom row (Figs. 8g–8i) shows the vapor quality contours.

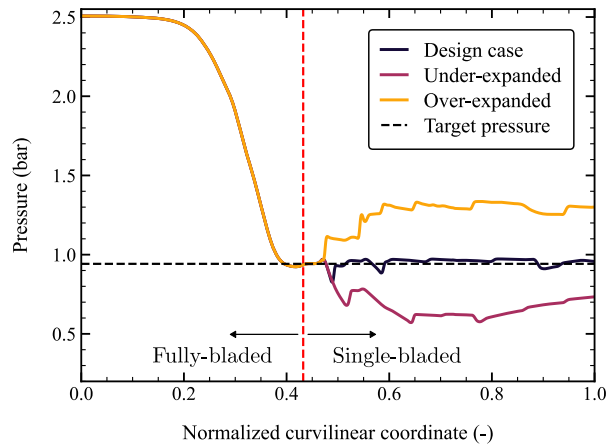


Fig. 10 Pressure distribution along the axis of a single stator passage for the design case and the under-expanded and over-expanded off-design conditions.

4.1.2 Sensitivity analysis on inlet temperature. The second sensitivity analysis investigates the effect of the heat source tem-

perature, varied from 80 to 150 °C in steps of 10 °C. The study is conducted for three inlet qualities: 0 %, 50 %, and 100 %. The main results are summarized in Fig. 8: (a) shows the expansion paths on the Cyclopentane T - s diagram, (b) presents the corresponding divergent nozzle geometries for each condition, and (c) depicts the Mach number distribution as a function of the normalized pressure drop.

By focusing on the impact of changing the inlet temperature level while keeping the same quality, for instance, for $Q_0 = 0$ %, it is possible to see a slight difference in nozzle shapes. The longest nozzles, for $Q_0 = 0$ % is the one at the lowest temperature, while for qualities of 50 % and 100 %, the longest is the one at the highest total temperature. The results indicate that changes in inlet total temperature, while maintaining constant total quality, have a comparatively smaller effect on the nozzle shape than variations in inlet quality (Fig. 8 (b)). Additionally, when visualizing the Mach number distributions with respect to the normalized pressure (Fig. 8 (c)), the curves corresponding to same inlet quality and different pressure levels tend to collapse, showing near to same pressure ratio behavior. The reason why the results are clustered by inlet quality is because $\rho_v \ll \rho_\ell$ for the temperature range of interest. Under this condition, the volume ratio depends mostly on the amount of liquid that evaporates compared to the initial amount of vapor, which is similar in all cases with the same inlet quality.

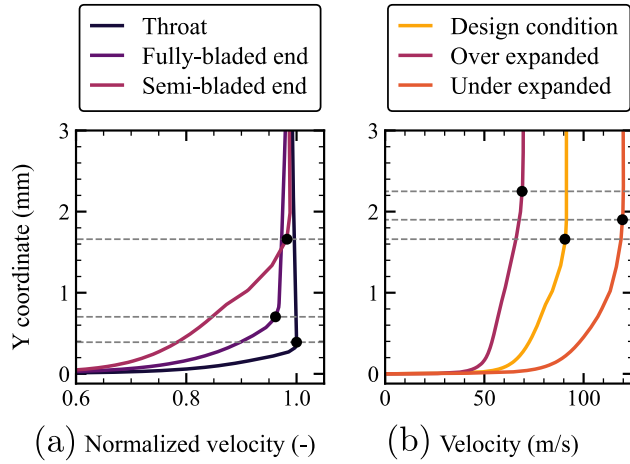


Fig. 11 (a) Boundary layer development at the throat, end of semi-bladed region, and end of fully-bladed region for the design case. (b) Boundary layer development at the end of the semi-bladed region for the design, over-expanded, and under-expanded cases. The boundary layer thickness is marked by a black dot (•).

4.2 Two-phase supersonic stator design. The stator blade parametrization introduced in Section 2.2 is applied to the nozzle geometry designed using the boundary conditions summarized in Table 3. The main geometrical features of the design are listed in Table 1.

4.2.1 Sensitivity analysis on back pressure. Three operating conditions are assessed through CFD simulations: the design point and two off-design points obtained by varying the outlet pressure by $\pm 25\%$ with respect to its nominal value. The reduced outlet pressure corresponds to an under-expanded regime, whereas the increased outlet pressure corresponds to an over-expanded one. Figure 9 illustrates the flow fields in terms of static pressure, Mach number, and vapor quality for the three cases. Fig. 10 reports the pressure distribution along the passage centerline. Figure 11 shows the boundary layer development for (a) the design conditions at three different locations and (b) for the design, under, and over-expanded cases at the end of the semi-bladed region. Figure 12 shows (a) the Mach number and (b) the flow angle pitch-wise distributions.

At the inlet, the flow is subsonic and aligned with the blade axial direction, and it is subsequently turned and accelerated through the converging portion of the passage. As the fluid expands, the static pressure decreases while the vapor quality rises. The sonic conditions are reached at the cascade throat, followed by a supersonic acceleration in the diverging section. Up to the end of the fully-bladed region, the flow field is essentially identical for all three operating conditions; the differing back-pressures only affect the downstream region past the trailing edge, as also illustrated in Fig. 10, where the end of the MoC-designed region, at the end of the fully-bladed section, is highlighted.

In the design case, the flow is well guided through the cascade, with only weak expansion waves emerging at the trailing edge. In the Mach contour of Fig. 9 (e), the wake has a thickness comparable to that of the trailing-edge, and remains straight and well aligned with the blade direction. As shown in Fig. 11 (a), the boundary-layer thickness increases progressively from the throat (approximately 0.5 mm) to the end of the fully bladed section (about 0.8 mm), and reaches roughly 1.6 mm at the end of the semi-bladed region. The pitchwise distributions of Mach number and flow angle reported in Fig. 12 confirm excellent agreement with the design targets, indicated by the black dashed lines. The Mach number distribution exhibits a characteristic dip associated

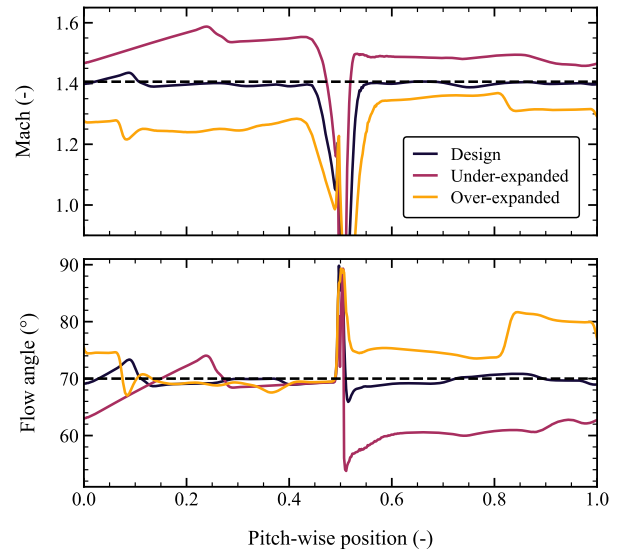


Fig. 12 Pitch-wise distributions downstream of the trailing edge of Mach number (top) and flow angle (bottom) for the three operating conditions.

with the wake region, marked by reduced velocity and increased dissipation, due to the proximity of the evaluation line to the trailing edge. The mass-weighted outlet Mach number computed from the CFD simulation for the design case is 1.38 (Table 3), deviating by 1.85 % from the design Mach number of 1.406. This demonstrates that the proposed MoC-based parametrization enables the accurate design of two-phase supersonic stator blades when the thickness of the boundary layer is small compared to the nozzle size.

In the under-expanded case, the diverging portion of the nozzle does not provide sufficient expansion to match the imposed exit pressure. As a result, an expansion fan originates downstream of the pressure-side trailing edge, causing an additional acceleration of the flow. This behavior is evident in the semi-bladed region of the centerline pressure distribution shown in Fig. 10, where a further pressure drop is observed. The additional turning is achieved through a supersonic expansion, reflected in the reduction of flow angle in the pitchwise distributions of Fig. 12. The resulting average exit flow angle is 63.64° (Table 3), more than 6° lower than the design value. Although the loss in efficiency remains small, with a reduction of 1.52% below the design point, the reduction in flow angle is substantial and may induce incidence losses on the downstream rotor, potentially resulting in performance penalties at the turbine stage level. The associated deviation is also visible in the Mach number contours of Fig. 9 (d), where the wake no longer aligns with the blade direction. The reflection of the expansion fan on the suction side increases the boundary-layer thickness at the exit of the single-blade passage by 14.46 % compared with the design case (Fig. 11 (b)).

Finally, in the over-expanded case, the pressure at the exit of the fully-bladed region falls below the imposed backpressure. As a consequence, the flow is compressed through an pair of oblique shocks that originate from the wake region, also known as fish-tail shocks (Fig. 9). These shocks propagate from the pressure side of the blade and reflect on the suction side of the adjacent blade, producing shock-boundary-layer interactions that thicken the boundary layer and increase viscous dissipation. The increase in boundary layer thickness is caused by a small pressure-driven separation bubble due to the reflecting shock on the suction side, visible in the Mach contours of Fig. 9 (f), resulting in a 35.54 % thickness increase relative to the design case (Fig. 11 (b)). As can be noticed in Fig. 9 (f), the complex shock system interacts

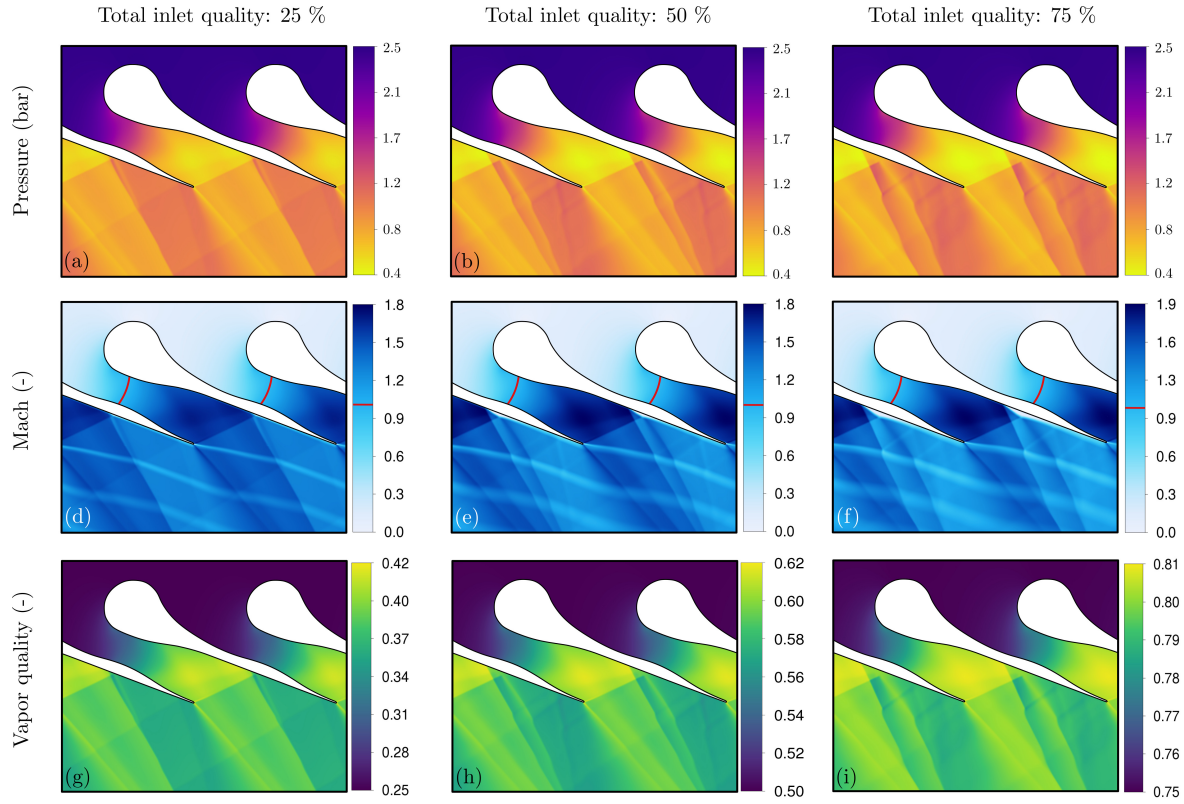


Fig. 13 Impact of varying total inlet quality on flow characteristics. Results are shown for total inlet qualities of 25 % (left column), 50 % (center column), and 75 % (right column). The top row (Figs. 13a–13c) shows pressure contours, the central row (Figs. 13d–13f) shows Mach contours, and the bottom row (Figs. 13g–13i) shows the vapor quality contours.

Table 4 Performance indicators for the sensitivity analysis on total inlet quality.

$Q_{0,in}$	25 %	50 %	75 %	Unit
v_{out}	144.78	185.15	217.74	m/s
$v_{out,is}$	153.71	201.19	239.44	m/s
α_{out}	73.86	74.65	74.94	°
Ma_{out}	1.34	1.28	1.25	—
η_{is}	88.71	84.68	82.69	%

with the wake region, which is not straight and more diffuse than in the under-expanded and design cases. Although the flow angle remains largely unaffected—reduced by only 2.36° relative to the design condition—the oblique shocks act as strong entropy-generation mechanisms, decreasing the exit velocity and reducing the isentropic efficiency by approximately 5.5 percentage points (Table 3).

4.2.2 Sensitivity analysis on inlet quality. Uncertainties regarding the power source can cause variations in the turbine inlet quality, potentially impacting aerodynamic performance. To investigate this, three off-design conditions with total inlet qualities of 25 %, 50 %, and 75 % are investigated, which represent significantly higher values than the design one of 2.5 %. The stagnation inlet pressure and the outlet static pressure are kept the same to the design case. The pressure, Mach number, and vapor quality contours for these off-design scenarios are presented in Fig. 13, while the design case contours are shown in Figs. 9 (b), (e), and (h). Additionally, Fig. 14 illustrates the pressure distribution along the passage centerline, and the primary quantitative flow features are reported in Table 4.

The results presented in Figures 13 and 14 indicate that all

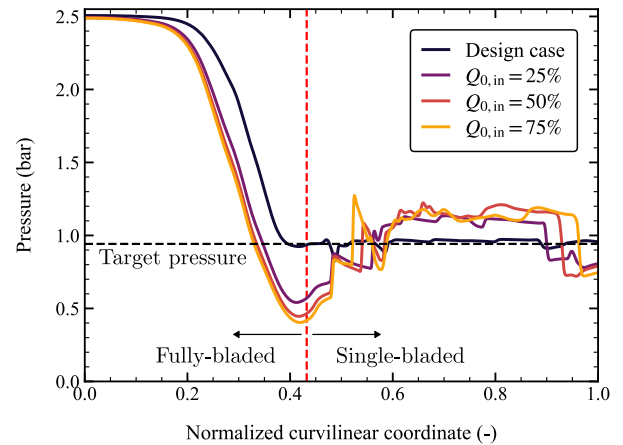


Fig. 14 Pressure distribution along the axis of the stator passage for three off-design values of total inlet quality.

three cases lead to flow overexpansion, followed by a complex shock wave pattern to match the imposed outlet pressure. This overexpansion is evident in the pressure distribution in Fig. 14, where minimum pressures reach approximately 0.5 bar, significantly lower than the outlet and target pressure of 0.95 bar. The three cases exhibit similar flow features: a shock wave generated on the suction side due to rapid expansion, and a second shock wave departing from the trailing edge that impinges on the suction side just after the departure of the first one. The shock strength increases with higher total inlet qualities (50 % and 75 %), as shown by the separation bubble on the suction side in the Mach

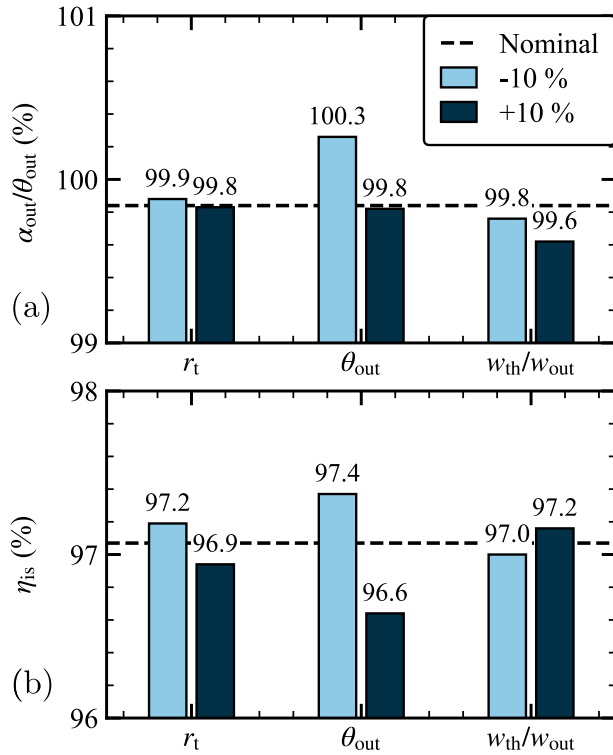


Fig. 15 Impact of trailing radius, outlet metal angle and throat-to-outlet opening ratio on (a) normalized outlet flow angle and (b) stator isentropic efficiency.

contours of Fig. 13. This separation originates at the first suction-side shock and is intensified by interaction with the trailing-edge shock. The magnitude of these shock waves is further detailed in Fig. 14, where the trailing-edge shock occurs at a normalized curvilinear coordinate of approximately 0.48 for all cases. Its intensity is comparable across the three scenarios and is followed by a brief expansion before the pressure distribution is affected by the shock wave originating from the trailing edge. The impact of this second shock is visible at normalized curvilinear coordinates of 0.55, 0.52, and 0.50 for inlet qualities of 25 %, 50 %, and 75 %, respectively. The 25 % quality case shows the lowest intensity, while 75 % quality case exhibits the maximum strength. According to the results presented in Table 4, the stator isentropic efficiency remains relatively high for all the cases, with drops of 8.59 %, 12.62 %, and 14.61 % compared to the design condition for the 25 %, 50 %, and 75 % cases. The most pronounced efficiency reduction is observed at higher vapor qualities. This behavior is consistent with the Mach number versus pressure trends shown in Fig. 8, where expansions originating from the liquid saturation line exhibit significantly different trajectories compared to lower inlet quality cases. These deviations indicate a substantial decrease in the expansion volumetric ratio relative to the design condition, which explains the observed performance degradation under the corresponding off-design operating points. Finally, the outlet flow angles in Table 4 are only slightly influenced by the shock structures, with a maximum deviation of 4.94° for the case with the highest inlet quality. This limited deviation suggests that rotor incidence angle effects may be negligible during off-design operation, though further CFD simulations involving a full turbine stage are required for confirmation.

4.2.3 Sensitivity analysis on geometric parameters. A sensitivity analysis is performed to assess the influence of the main geometrical parameters on stator performance. The three geometrical parameters considered are the trailing-edge radius, the outlet

metal angle, and the throat-to-outlet width ratio. The trailing-edge radius affects the flow perturbation at the stator exit and the strength of the fish-tail shock wave originating from the trailing edge. The outlet metal angle governs the flow turning, while the throat-to-outlet width ratio establishes the exit flow Mach number and can be modified from the MoC design by imposing a different target Mach number at the nozzle exit.

All design variables are varied by ± 10 % with respect to the nominal configuration, whose geometrical parameters are reported in Table 1. Specifically, the trailing-edge radius r_t is varied from 0.45 to 0.55 mm, the outlet metal angle θ_{out} from 63° to 77°, and the width ratio w_{th}/w_{out} from 0.54 ($Ma_{out} = 1.45$) to 0.66 ($Ma_{out} = 1.35$). The normalized outlet flow angle with respect to the outlet metal angle and the stator isentropic efficiency are reported in Figure 15.

Among the investigated parameters, the outlet metal angle has the most pronounced impact on performance, with an efficiency increase of 0.30 % when the outlet metal angle is reduced by 10 %, and a decrease of 0.43 % when it is increased by 10 %. This behavior is likely associated with changes in wetted area, which affect viscous losses and, consequently, the stator isentropic efficiency. The second most influential parameter is the trailing-edge radius, which leads to an efficiency increase of 0.12 % when reduced and a decrease of 0.13 % when increased by 10 %. The throat-to-outlet width ratio has a positive impact of 0.10 % when the outlet opening is reduced, due to the weakening of the fish-tail shock wave at lower exit Mach numbers. Conversely, a 10 % reduction in the ratio w_{th}/w_{out} results in an efficiency decrease of 0.07 %. Overall, variations in both the outlet flow angle and the isentropic efficiency remain negligible, highlighting the robustness of the MoC extended to two-phase flows for the design of efficient supersonic stators operating at design conditions.

5 Conclusions

This work presented an extension of the Method of Characteristics (MoC) for designing two-phase convergent-divergent nozzles, which are relevant to applications such as waste heat recovery and liquefaction systems. Furthermore, the proposed approach was integrated into a parametrized design framework for supersonic stator blades in two-phase turbines. The main findings of the study are summarized below.

- The barotropic flow formulation demonstrates to reliably capture non-ideal gas effects in supersonic regimes. Comparison between numerical predictions and the experimental data of Spinelli et al. [39] confirms the model accuracy, with Normalized Root Mean Square Errors (NRMSE) below 2.75 %.
- The extended MoC is verified for both single-phase and two-phase expanding flows, with excellent agreement with 2D CFD in axial distributions and iso-Mach contours, yielding a critical mass-flow deviation below 0.05 % and NRMSE below 0.7 %.
- Sensitivity analyses on total inlet quality and temperature highlights the strong influence of inlet thermodynamic conditions on the nozzle geometry. The void fraction distribution, which governs the two-phase expansion, emerged as a key parameter controlling the shape and performance of the nozzle.
- The divergent section generated through the Method of Characteristics was successfully integrated into a fully parametrized supersonic stator cascade. High-fidelity CFD simulations demonstrated a smooth and shock-free expansion under design conditions, yielding a uniform exit flow at the target Mach number and flow angle. The design achieved an isentropic efficiency of 97.30 %, with minimal degradation under off-design conditions. A reduction of only 1.52 percentage points in the under-expanded case and 5.62 percentage points in the over-expanded condition is calculated, the

latter attributed to the presence of stronger shock waves. Furthermore, the designed stator blades demonstrated robustness against variations in total inlet quality while maintaining relatively high efficiency. This confirms the reliability of the proposed design despite potential uncertainties in the power source.

The proposed method establishes a robust and computationally efficient framework for designing two-phase nozzles and turbine stator cascades. Compared to conventional optimization-based design approaches, it provides direct physical insight and significantly reduces computational cost, while maintaining high predictive accuracy. This contribution advances the development of high-performance two-phase turbines, supporting ongoing efforts in the decarbonization of thermal energy conversion technologies.

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Nomenclature

Acronyms

CFD	Computational fluid dynamics
EoS	Equation of state
HEM	Homogeneous equilibrium model
HEOS	Helmholtz equation of state
MDM	Octamethyltrisiloxane
MoC	Method of characteristics
NRMSE	Normalized root mean square error
ODE	Ordinary differential equation
ORC	Organic Rankine cycle
PDE	Partial differential equation
RANS	Reynolds-averaged Navier–Stokes
RGP	Real-gas property

Roman letters

a	Sound speed (m/s)
c	Specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)
CP	Control point
h	Enthalpy (J kg^{-1})
H	Height (m)
Ma	Mach number (-)
P	Pressure (Pa)
Q	Vapor quality (-)
r	Radius (m)
s	Entropy or blade pitch ($\text{J kg}^{-1} \text{K}^{-1}$ or m)
T	Temperature (K)
u, v	Velocity components (m/s)
w	Channel width (m)
x	Cartesian coordinate (m)
y	Cartesian coordinate / mass fraction (-)

Greek letters

α	Mach angle / volume fraction (-)
δ	Axisymmetric factor
η	Efficiency (-)
ρ	Density (kg m^{-3})
θ	Flow angle (rad)

Superscripts and subscripts

0	Stagnation conditions
$\pm$	Left- and right-running characteristics
in	Inlet
is	Isentropic
l	Leading edge
ℓ	Liquid phase
out	Outlet
ref	Reference
sat	Saturation
sim	Simulated
t	Trailing edge
th	Throat
v	Vapor phase
x, y	Spatial partial derivatives

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List of Figures

1	Mixture speed of sound as a function of vapor quality under different equilibrium assumptions for cyclopentane at a saturation temperature of 200 °C, calculated using the expressions derived in [30]. Under full thermodynamic equilibrium, the speed of sound exhibits a discontinuous drop upon entering the two-phase region, which is especially pronounced at the liquid saturation line and much weaker near saturated vapor.	2
2	Parametrization of the stator blades.	4
3	Stator mesh with three zoom regions: (a) leading edge, (b) throat, and (c) trailing edge.	5
4	Validation of the barotropic model for real gas with the experiments from Spinelli et al.[39]. (a) Pressure and (b) compressibility factor distributions along the nozzle axis.	6
5	Validation of the method of characteristics for single-phase flows. (a) Wall and (b) axis Mach and pressure distributions.	6
6	Comparison between the Mach lines obtained from the MoC design and two-dimensional CFD simulations employing the barotropic model for total inlet qualities of (a) 2.5 % and (b) 25 %.	6
7	Sensitivity analysis of total inlet quality variations for a fixed nozzle exit Mach number of 1.50. (a) Expansion processes on the Cyclopentane $T-s$ diagram; (b) corresponding nozzle geometries obtained using the MoC; (c) isentropic Mach number evolution as a function of the normalized pressure P/P_0	7
8	Sensitivity analysis of total inlet quality and temperature variations for a fixed nozzle exit Mach number of 1.50: (a) Expansion paths on the Cyclopentane $T-s$ diagram; (b) corresponding nozzle geometries designed using the MoC; (c) isentropic Mach number evolution as a function of the normalized pressure P/P_0	7
9	Outlet pressure is varied to obtain three flow conditions: under-expanded ($P_{out} < P_{design}$, first column), ideally expanded ($P_{out} = P_{design}$, second column), and over-expanded ($P_{out} > P_{design}$, third column). The top row (Figs. 8a–8c) shows pressure contours, the central row (Figs. 8d–8f) shows Mach contours, and the bottom row (Figs. 8g–8i) shows the vapor quality contours.	8
10	Pressure distribution along the axis of a single stator passage for the design case and the under-expanded and over-expanded off-design conditions.	8
11	(a) Boundary layer development at the throat, end of semi-bladed region, and end of fully-bladed region for the design case. (b) Boundary layer development at the end of the semi-bladed region for the design, over-expanded, and under-expanded cases. The boundary layer thickness is marked by a black dot (•).	9
12	Pitch-wise distributions downstream of the trailing edge of Mach number (top) and flow angle (bottom) for the three operating conditions.	9
13	Impact of varying total inlet quality on flow characteristics. Results are shown for total inlet qualities of 25 % (left column), 50 % (center column), and 75 % (right column). The top row (Figs. 13a–13c) shows pressure contours, the central row (Figs. 13d–13f) shows Mach contours, and the bottom row (Figs. 13g–13i) shows the vapor quality contours.	10
14	Pressure distribution along the axis of the stator passage for three off-design values of total inlet quality.	10
15	Impact of trailing radius, outlet metal angle and throat-to-outlet opening ratio on (a) normalized outlet flow angle and (b) stator isentropic efficiency.	11

List of Tables

1	Stator geometrical parameters.	4
2	Table	5
3	Boundary conditions and key performance indicators for the two-phase supersonic stator simulations.	7
4	Performance indicators for the sensitivity analysis on total inlet quality.	10